

Review that CanvasNet Drawing Subsystem Implementation is Correct

2026-10-02

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1 Review that CanvasNet Drawing Subsystem Implementation is Correct

1.1 1. Introduction

1.1.1 1.1 Purpose

This document records the formal review of a set of project files.

1.1.2 1.2 Scope

This is the `CanvasNet-Drawing` ReviewMark review-set: a subsystem-level review proving that the `Drawing` subsystem's top-level design and verification documentation accurately and completely describe what its constituent units provide, and that the subsystem requirements are satisfied — per the hierarchical-scope principle, unit-level implementation detail (source code, tests, and unit design/verification/requirements documents) is explicitly excluded and relied upon only through the subsystem's own description of it. This effort added a new unit, `TilePaint`, to the `Drawing` subsystem (reviewed separately in its own `CanvasNet-Drawing-TilePaint` unit-level review-set) and updated `docs/reqstream/canvas-net/drawing.yaml` to add the corresponding subsystem-level requirement/cross-reference. This is the first formal review ever conducted for this review-set in this repository (the `reviews` branch `index.json` was empty prior to this review).

1.1.3 1.3 Outcomes

Each check must be recorded with one of the following outcomes:

Outcome	Meaning
Pass	The check was performed and the criterion is satisfied
Fail	The check was performed and the criterion is not satisfied
N/A	The check does not apply; justification is required

1.1.4 1.4 Review Details

Field	Value
Project	CanvasNet
Review ID	CanvasNet-Drawing
Review Title	Review that CanvasNet Drawing Subsystem Implementation is Correct
Fingerprint	5d671b27755d69066b2ce3d64ab02712c034cf141921ecc1e43a44b58017e1ba
Review Date	2026-10-02

1.1.5 1.5 Reviewers

Name	Role	Organization	Signature	Date
Formal Review Agent	Automated Formal Reviewer	DEMA Consulting	(agent-generated, no handwritten signature)	2026-10-02

1.1.6 1.6 Files Under Review

File
docs/reqstream/canvas-net/drawing.yaml
docs/design/canvas-net/drawing.md
docs/verification/canvas-net/drawing.md

Context (not under review, for orientation only): docs/design/canvas-net.md , docs/reqstream/canvas-net.yaml , docs/design/introduction.md .

1.2 2. Review Checklist

1.2.1 2.1 Requirements Checks

Applicable: Yes

#	Check	Outcome	Justification
REQ-01	All requirements have a unique identifier	Pass	All IDs in <code>drawing.yaml</code> (e.g. <code>CanvasNet-Drawing-TilePaint</code> , <code>CanvasNet-Drawing-TilePaintValidation</code>) are unique within the file.
REQ-02	All requirements are unambiguous	Pass	Each requirement statement is a single, clearly worded sentence.
REQ-03	All requirements are testable	Pass	Each requirement lists concrete <code>tests:</code> entries (e.g. <code>PathFiller_Fill_TilePaint_CheckerboardTile_FillsWithBothTileColors</code>).
REQ-04	All requirements are consistent	Pass	No contradictions found between <code>TilePaint / TilePaintValidation</code> and the pre-existing <code>GradientPaint / GradientPaintValidation</code> requirements they were modeled after.
REQ-05	All requirements are complete (no TBDs)	Pass	No placeholders or TBDs present.
REQ-06	All requirements are verifiable	Pass	Requirements describe objectively observable rendering/validation outcomes.
REQ-07	No compound requirements are present	Pass	<code>CanvasNet-Drawing-TilePaint</code> and <code>CanvasNet-Drawing-TilePaintValidation</code> each express one coherent testable criterion (the validation requirement groups three argument-null checks sharing one mechanism, consistent with the existing <code>GradientPaintValidation / PathFillValidation</code> pattern in this same file).
REQ-08	No requirements are missing	Fail	The file's top comment and requirement set correctly describe the <code>TilePaint</code> unit's addition, but the subsystem's own design and verification documents (the other two files in this review-set) do not reflect the new requirements at all — see DES-02, REQDOC-01, and the Summary of Findings. This is a documentation-completeness gap triggered by, not located in, the requirements file; see 3.1.

1.2.2 2.2 Design Documentation Checks

Applicable: Yes

#	Check	Outcome	Justification
DES-01	Design documentation clearly describes the purpose of the component or feature	Fail	<code>drawing.md</code> 's Purpose section states the subsystem "currently contains three sibling units" and enumerates only <code>PathFiller</code> , <code>PathStroker</code> , and <code>GradientPaint</code> ; it omits <code>TilePaint</code> entirely, so the stated purpose no longer matches the subsystem's actual unit composition.
DES-02	Design documentation covers the necessary implementation details	Fail	The Units section lists only three units; there is no <code>TilePaint</code> bullet (compare the existing <code>GradientPaint</code> bullet, which correctly cross-references <code>_GradientPaint Unit Design_</code> (<code>drawing/gradient-paint.md</code>) — no equivalent exists for <code>_TilePaint Unit Design_</code> (<code>drawing/tile-paint.md</code>) , even though that unit design document exists on disk).
DES-03	Design documentation describes how the code is interfaced (APIs, inputs, outputs)	Fail	<code>TilePaint</code> 's public surface and its consumption via <code>PathFiller</code> 's tile-paint <code>Fill</code> overload are not described anywhere in this document, unlike the equivalent description given for <code>GradientPaint</code> 's gradient <code>Fill</code> overload.
DES-04	Design documentation describes the expected normal operation	Pass	For the three units it does describe (<code>PathFiller</code> , <code>PathStroker</code> , <code>GradientPaint</code>), normal-operation collaboration is clearly described.
DES-05	Design documentation describes the expected error handling	N/A	Subsystem-level design intentionally defers error-handling detail to unit design documents per <code>design-documentation.md</code> ; the existing text does not describe error handling for any unit (including the three it does cover), so this is a pre-existing, uniform omission rather than a <code>TilePaint</code> -specific regression.

1.2.3 2.3 Technical Documentation Checks

Applicable: No

No general technical documentation files (user guides, API references, README, release notes) are in this review-set's file list; `drawing.md` and `drawing.yaml` are design/requirements artifacts covered by sections 2.2 and 2.1 respectively.

1.2.4 2.4-2.10, 2.12 Code/Test/Security/Readability Checks

Applicable: No

This subsystem-level review-set contains no source code or test files per the hierarchical-scope principle (`.reviewmark.yaml` deliberately limits `CanvasNet-Drawing` 's paths to the subsystem requirements, design, and verification documents only); unit source and tests are reviewed under the separate `CanvasNet-Drawing-TilePaint` , `CanvasNet-Drawing-PathFiller` , `CanvasNet-Drawing-PathStroker` , and `CanvasNet-Drawing-GradientPaint` unit-level review-sets.

1.2.5 2.11 Requirements vs Documentation Checks

Applicable: Yes

Both requirements (`drawing.yaml`) and design/verification documentation (`drawing.md` x2) are in this review-set's file list.

#	Check	Outcome	Justification
REQDOC-01	All reviewed requirements are addressed in the general technical documentation	Fail	<code>drawing.yaml</code> 's <code>CanvasNet-Drawing-TilePaint</code> and <code>CanvasNet-Drawing-TilePaintValidation</code> requirements have no corresponding coverage in <code>docs/design/canvas-net/drawing.md</code> (<code>Purpose/Units/Dependencies/Callers</code> all omit <code>TilePaint</code>) or in <code>docs/verification/canvas-net/drawing.md</code> (<code>Verification Approach and Acceptance Criteria</code> both omit <code>_TilePaint Unit Verification Design_</code>). All other requirements in the file (<code>PathFill</code> , <code>PathStroke</code> , <code>GradientPaint</code> , and their validation/no-op variants) are correctly addressed in both documents.
REQDOC-02	No reviewed requirement is contradicted by the general technical documentation	Pass	For the requirements that are addressed, no contradictions were found; the gap is one of omission (REQDOC-01), not contradiction.

1.2.6 2.13 Requirements vs Testing Checks

Applicable: Yes

The requirements file links each requirement to named tests; no test source files are in this review-set's scope, so applicability is limited to confirming traceability linkage is present.

#	Check	Outcome	Justification
REQTEST-01	Every requirement under review is covered by at least one test	Pass	<code>CanvasNet-Drawing-TilePaint</code> links to <code>PathFiller_Fill_TilePaint_CheckerboardTile_FillsWithBothTileColors</code> and <code>EvaluateRow_WrapAroundBeyondOneTile_MatchesFirstTileRepetition</code> ; <code>CanvasNet-Drawing-TilePaintValidation</code> links to three named null-argument tests. Every other requirement in the file also has a non-empty <code>tests:</code> list. Actual test pass/fail status is unit-level verification evidence, outside this subsystem review-set's scope.
REQTEST-02	Tests verify the behavior described in each requirement	Pass	Test names read as direct paraphrases of their requirement's stated behavior (e.g. <code>TilePaint</code> sampling/wrap-around behavior, null-argument rejection).

1.2.7 2.14 Code vs Design Documentation Checks

Applicable: No

No source code files are in this review-set's scope (hierarchical-scope principle); code-to-design consistency for `TilePaint`'s actual implementation is assessed under the `CanvasNet-Drawing-TilePaint` unit-level review-set.

1.3 3. Conclusion

1.3.1 3.1 Summary of Findings

#	Check	Finding
REQ-08	No requirements are missing	Symptom of the same gap as DES-01/02/03 and REQDOC-01: the requirements file is itself complete and correctly updated for <code>TilePaint</code> , but the subsystem design/verification documents were not updated to match, leaving the requirement set effectively undocumented at the subsystem level.
DES-01	Purpose describes the component	<code>docs/design/canvas-net/drawing.md</code> 's Purpose section states the subsystem "currently contains three sibling units" (<code>PathFiller</code> , <code>PathStroker</code> , <code>GradientPaint</code>) with no mention of <code>TilePaint</code> , even though <code>TilePaint</code> is a fourth sibling unit added alongside this change.
DES-02 / DES-03	Design covers necessary implementation detail / interfaces	The Units section of the same document has no <code>TilePaint</code> bullet (unlike the parallel <code>GradientPaint</code> bullet), so it does not cross-reference <code>_TilePaint Unit Design_</code> (<code>drawing/tile-paint.md</code>) , does not describe <code>TilePaint</code> 's public surface, and the Dependencies/Callers sections likewise never mention <code>TilePaint</code> or its internal <code>TilePaintEvaluator</code> helper.
REQDOC-01	Requirements vs documentation	<code>docs/verification/canvas-net/drawing.md</code> has the identical gap on the verification side: its intro, Verification Approach, and Acceptance Criteria sections list <code>PathFiller</code> , <code>PathStroker</code> , and <code>GradientPaint</code> unit verification designs only — <code>_TilePaint Unit Verification Design_</code> (<code>drawing/tile-paint.md</code>) is never cited, so the subsystem's stated acceptance criteria do not include <code>TilePaint</code> 's test scenarios passing.
Observation (informational, out of	Cross-file consistency	The parent system design document <code>docs/design/canvas-net.md</code> (global context, not a file under review here) has the same staleness one

#	Check	Finding
this review-set's scope)		level up: its <code>Drawing subsystem</code> bullet enumerates “three public entry points” / three units and its later collaboration narrative says “The <code>Drawing</code> subsystem’s two units collaborate as follows” — both predate <code>GradientPaint</code> as well as <code>TilePaint</code> and are unaffected by this review’s pass/fail determination, but should be corrected in the <code>CanvasNet-Design</code> / <code>CanvasNet-Architecture</code> review-sets, which do include <code>canvas-net.md</code> as a file under review.
Positive finding	Requirements completeness	<code>docs/reqstream/canvas-net/drawing.yaml</code> was itself correctly and completely updated: its top-of-file comment names <code>TilePaint</code> as a covered unit, and it adds two well-formed, testable, non-contradictory requirements (<code>CanvasNet-Drawing-TilePaint</code> , <code>CanvasNet-Drawing-TilePaintValidation</code>) with correct downward links to unit-level requirement IDs and named tests.
Positive finding	SysML2 cross-check	The authoritative SysML2 model (<code>docs/sysml2/model/canvas-net/drawing.sysml</code> , consulted as orientation context) already declares <code>part tilePaint : TilePaint;</code> under the <code>Drawing</code> part def and describes <code>TilePaint</code> in the part def’s <code>doc</code> comment — confirming the structural model and the unit’s own artifacts (<code>docs/design/canvas-net/drawing/tile-paint.md</code> , <code>docs/verification/canvas-net/drawing/tile-paint.md</code> , <code>docs/reqstream/canvas-net/drawing/tile-paint.yaml</code> , and the <code>TilePaint.cs</code> / <code>TilePaintEvaluator.cs</code> source) all exist and are complete. The sole gap is that the two subsystem-level roll-up documents under review here were never updated to reflect them.

1.3.2 3.2 Overall Outcome

Overall Outcome: Fail

The `Drawing` subsystem's requirements (`docs/reqstream/canvas-net/drawing.yaml`) were correctly and completely updated to add the new `TilePaint` unit. However, both of the subsystem's other mandatory top-level documents under review — the design document (`docs/design/canvas-net/drawing.md`) and the verification design document (`docs/verification/canvas-net/drawing.md`) — were **not** updated: both still describe the `Drawing` subsystem as containing exactly three sibling units (`PathFiller` , `PathStroker` , `GradientPaint`) and omit every mention of `TilePaint` , its unit design document, its unit verification document, and its role in the subsystem's dependency/caller/acceptance-criteria narrative. This fails the specific compliance question this review-set exists to prove: whether the `Drawing` subsystem's top-level design and verification documentation still accurately and completely describe what its units provide and that its own subsystem requirements are satisfied.

Required follow-up before this review can be considered approved:

1. Update `docs/design/canvas-net/drawing.md` 's Purpose section to state four sibling units (not three) and add a `TilePaint` bullet to the Units section, following the existing `GradientPaint` bullet's pattern, cross-referencing `_TilePaint Unit Design_` (`drawing/tile-paint.md`) .
2. Update the same document's Dependencies and Callers sections to describe `TilePaint` 's dependency on `PathFiller` 's tile-paint `Fill` overload and its internal `TilePaintEvaluator` helper, consistent with how `GradientPaint` / `GradientEvaluator` are already described.
3. Update `docs/verification/canvas-net/drawing.md` 's intro, Verification Approach, and Acceptance Criteria sections to cite `_TilePaint Unit Verification Design_` (`drawing/tile-paint.md`) alongside the existing three unit verification design references.
4. Re-run this review after the above corrections are made.